$$\frac{}{\Gamma \vdash n : \mathsf{num}}\ \text{T-Num} \qquad \frac{}{\Gamma \vdash b : \mathsf{bool}}\ \text{T-Bool} \qquad \frac{\Gamma \vdash e_1 : \tau_1 \quad \Gamma \vdash e_2 : \tau_2}{\Gamma \vdash \langle e_1, e_2 \rangle : \langle \tau_1, \tau_2 \rangle}\ \text{T-Tuple}$$

$$\frac{}{\Gamma \vdash \mathsf{unit} : \mathsf{unit}}\ \text{T-Unit} \qquad \frac{o : \tau_a \times \tau_b \to \tau_o \quad \Gamma \vdash e_1 : \tau_a \quad \Gamma \vdash e_2 : \tau_b}{\Gamma \vdash e_1\ o\ e_2 : \tau_o}\ \text{T-Op}$$

$$\frac{x : \tau \in \Gamma}{\Gamma \vdash x : \tau}\ \text{T-Var} \qquad \frac{\Gamma, x : \tau_1 \vdash e : \tau_2}{\Gamma \vdash \lambda x : \tau_1 . e : \tau_1 \to \tau_2}\ \text{T-Lambda}$$

$$\frac{\Gamma \vdash e_1 : \tau_1 \to \tau_2 \quad \Gamma \vdash e_2 : \tau_1}{\Gamma \vdash e_1\ e_2 : \tau_2}\ \text{T-App} \qquad \frac{\Gamma \vdash e_1 : \mathsf{bool} \quad \Gamma \vdash e_2 : \tau \quad \Gamma \vdash e_3 : \tau}{\Gamma \vdash \text{if } e_1 \text{ then } e_2 \text{ else } e_3 : \tau}\ \text{T-If}$$

$$\frac{\Gamma \vdash e : \mathsf{num}}{\Gamma \vdash \mathsf{zero?}(e) : \mathsf{bool}}\ \text{T-Zero}$$

$$\frac{\Gamma \vdash e : \langle \tau_1, \tau_2 \rangle}{\Gamma \vdash \mathsf{inj.l}(e) : \tau_1}\ \text{T-Inj-Left} \qquad \frac{\Gamma \vdash e : \langle \tau_1, \tau_2 \rangle}{\Gamma \vdash \mathsf{inj.r}(e) : \tau_2}\ \text{T-Inj-Right}$$

Figure 1: Type Rules for Base